BASIC ROBOT MOVEMENTS

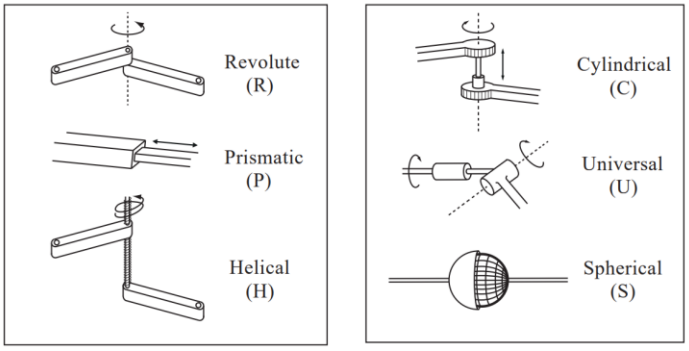
Robot movement is a concept commonly used in robotics, mechanics, and physics to describe the number of independent ways in which an object can move or change its configuration. An example of this is the actuation of the joints. Utilizing electric motors enables the robot to generate controlled movements and apply forces as needed.

Basic Terminologies

- A **link** refers to a rigid component of a robot that connects two or more joints. A link can be like a segment or an arm of a robot, which may have a fixed or variable length.
- A **joint** refers to the mechanism that connects two adjacent links in a robot, allowing them to rotate or translate relative to each other.
- A **robot's configuration** is a complete specification of the positions and orientations of each link of a robot, and its configuration space is the set of all possible configurations of the robot.
  - A person may specify the object's location as a set of (x, y, z) coordinates in some frame of reference. These are the Euler (pronounced "oiler") angles, φ, θ, γ, also known as pitch, yaw, and roll.
- **Degrees of freedom (DOF)** is the minimum number of independent parameters required to specify the position and orientation of each link based on the robot's configuration.

Robot Joints

- **Revolute joint R** (hinge joint) allows for rotational motion about the joint axis. (1 DOF)
- **Prismatic joint P** (sliding or linear joint) allows for translational (or rectilinear) motion along the direction of the joint axis. (1 DOF)
- **Screw joint H** (helical joint) allows simultaneous rotation and translation about a screw axis. (1 DOF)
- **Cylindrical joint C** allows for independent translations and rotations about a single fixed joint axis. (2 DOF)

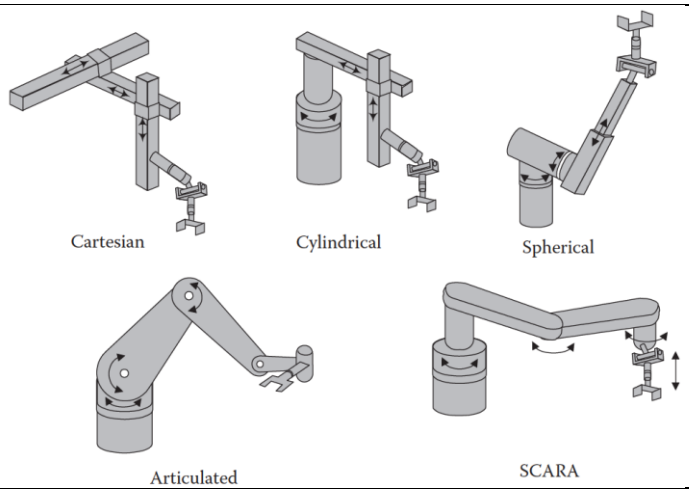


- **Universal joint U** consists of a pair of revolute joints arranged in which their joint axes are orthogonal. (2 DOF)
- **Spherical joint S** (ball-and-socket joint) has three degrees of freedom and functions like our shoulder joint. (3 DOF)

Robot Coordinates

It refers to a system of defining and measuring the position and orientation of a robot in each space or environment.

- **Cartesian/rectangular/gantry (3P)**: These robots use three prismatic joints to position the end effector, usually followed by additional revolute joints that orient the end effector.
- **Cylindrical (PRP)**: Cylindrical coordinate robots have two prismatic joints, one revolute joint for positioning the part and revolute joints for orientation.
- **Spherical (P2R)**: Spherical coordinate robots follow a spherical coordinate system with one prismatic and two revolute joints for positioning the part, plus additional revolute joints for orientation.



- **Articulated/anthropomorphic (3R)**: An articulated robot's joints are all revolute, like a human's arm. They are the most common configuration for industrial robots.
- **Selective Compliance Assembly Robot Arm (SCARA)**: SCARA robots have two (or three) revolute joints that are parallel and allow the robot to move in a horizontal plane, plus an additional prismatic joint that moves vertically. SCARA robots are very common in assembly operations.

Robot Reference Frames

Robots may be moved relative to different coordinate frames as follows, resulting in different motions.

- **World reference frame**
  This is a universal coordinate frame defined by x-, y-, and z-axes. In this case, the joints of the robot move simultaneously in a coordinated manner to create motions along the three major axes.
- **Joint reference frame**
  This is used to specify the movements of individual joints of the robot. In this case, each joint is accessed and moved individually; therefore, only one joint moves at a time.
- **Tool reference frame**
  This specifies movements of the robot's hand relative to a frame attached to the hand; consequently, all motions are relative to this local n, o, a-frame. Unlike the universal World frame, the local Tool frame moves with the robot.

Programming Modes

Robots may be programmed in several different modes, depending on the robot's complexity.

- **Physical setup**: In this mode, an operator sets up switches and hard stops that control the motions of the robot. This mode is usually used along with other devices, such as programmable logic controllers (PLCs).
- **Lead-through or teach mode**: In this mode, the robot's joints are moved with a teach pendant. When the desired location and orientation are achieved, the location is entered (taught) into the controller.

- **Continuous walk-through mode**: In this mode, all robot joints are moved simultaneously while the motion is continuously sampled and recorded by the controller.
- **Software mode**: In this mode, a program is written offline or online and is executed by the controller to control the motions. The programming mode is the most sophisticated and versatile mode and can include sensory information, conditional statements, and branching.

Levels of Robotic Languages

- **Microcomputer machine language level**: At this level, programs are written in machine language. This level of programming is the most basic and is very efficient, but it is difficult to understand and difficult for others to follow.
- **Point-to-point level**: At this level, the coordinates of the points are entered sequentially, and the robot follows the points as specified. This is a very primitive and simple type of program and is easy to use but not very powerful.
- **Primitive motion level**: In these languages, it is possible to develop more sophisticated programs, including sensory information, branching, and conditional statements. Most languages at this level are interpreter-based.
- **Structured programming level**: Most languages at this level are compiler-based, are powerful, and allow more sophisticated programming. However, they are also more difficult to learn.
- **Task-oriented level**: There are no actual languages in existence at this level – yet. Instead of programming a robot to perform a task by programming every step, the user would only mention the task, expecting that the controller would create the necessary sequence.

Robot Applications

Robots are best suited for applications involving repetitive tasks, jobs requiring capabilities and precision beyond those of humans, and working in hazardous environments. Robots have already been used in many industries and for many purposes.

- **Machine loading**. Robots supply other machines with parts or remove the processed parts from other machines.
- **Pick and place operations**. The robot picks up parts and places them elsewhere. This may include palletizing, placement of cartridges, simple assembly where two parts are put together (such as placing tablets into a bottle), placing parts in an oven, removing the treated parts from the oven, or other similar routines.
- **Welding**. The robot, along with proper setups and a welding end effector, is used to weld parts together. This is one of the most common applications of robots in the auto industry. Due to their consistent movements, robotic welds are very uniform and accurate.
- **Painting** is another very common application of robots, especially in the automobile industry. Since maintaining a ventilated but clean paint area suitable for humans is difficult to achieve, and because compared to humans' robotic operations are more consistent, painting robots are very well-suited for their job.
- **Inspection of parts** is also a very common application for robots. In general, robots are one component of an inspection system that may include a vision system, an X-ray device, an ultrasonic detector, or other similar devices.

- **Assembly tasks** usually involve many operations. Many of the fitting and assembling tasks are complicated and may require pushing, turning, bending, wiggling, pressing, snapping tabs to connect the parts, and other operations.
- **Manufacturing** may include many different operations, such as material removal, drilling, deburring, laying glue, cutting, and so on. It also includes the insertion of parts, such as electronic components into circuit boards, installation of boards into electronic devices, and other similar operations. Insertion robots are very common and are extensively used in the electronic industry. Figure 1.14 shows a robot engaged in manufacturing.
- **Medical applications** are also becoming increasingly common, such as surgical robots, which are not really robots because they are operated by a surgeon rather than operating on their own. No robot knows how to perform surgery yet. These precision robots assist surgeons in performing surgery.
- **Assisting disabled individuals** has also been tried with interesting results. Robots are well suited to help the disabled and elderly in many ways, and many humanoid robots under development can be excellent choices for this purpose. In addition, many exoskeleton devices have been developed to help paraplegic persons walk.
- **Hazardous environments**. Because of their inherent danger in these environments, humans must be well-protected. However, robots can access, traverse, maintain, and explore these areas without the same level of concern.
- **Inaccessible locations**, such as space, underwater, and mines, can also be serviced or explored by robots. Examples of these are underwater robots, which have also looked for, found, and explored sunken ships in deep oceans.
- **Robots in agriculture** can be very useful, especially in places where labor shortage or environmental conditions are important issues. Robotic pickers have been developed and tested around the world and are becoming more common.

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